

Question No 1:-

Part (a) :-

Exception:-

In programming, a program does not execute if it encounters an error. It can be a syntax error or any other error. In order to run our program and ignore these errors we handle exceptions by giving try and except statements.

Examples

General program for handling exception

Program:

try:

give something.

except (Value Error, Arithmetic Error):

handle multiple exceptions.

• IO Error:-

It is the type of error that occurs when an Input Output operation does not execute. When you open a file that does not exist.

Example:

try:

```
file = open("fizza.txt", "r")
```

except IO Error:

```
print "IO Error"
```

Arithmetic Error:

Arithmetic Error occurs when division by zero is given, it gives an error zero Division Error, floatingpoint error.

Index Error:

Index error is one of the most basic error. It occurs when we give wrong index that is outside of a list, string etc.

Example:-

```
list = ["fan", "man", "pan", "van"]
```

```
print(list[5])
```

Output:-

Index Error; List index out of range

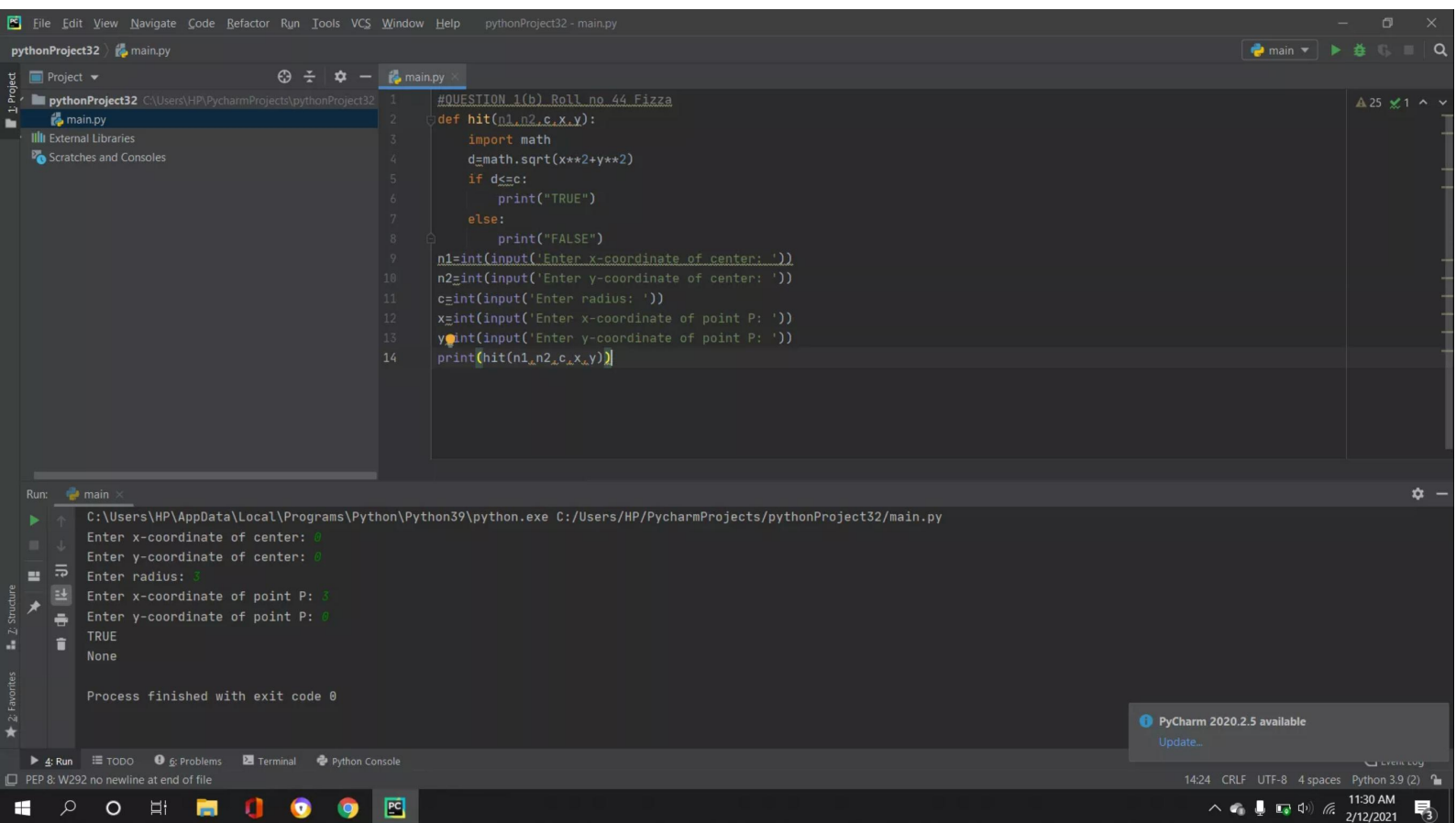
Value Error:- Value Error raise when a function receives an argument of the correct type but an inappropriate value.

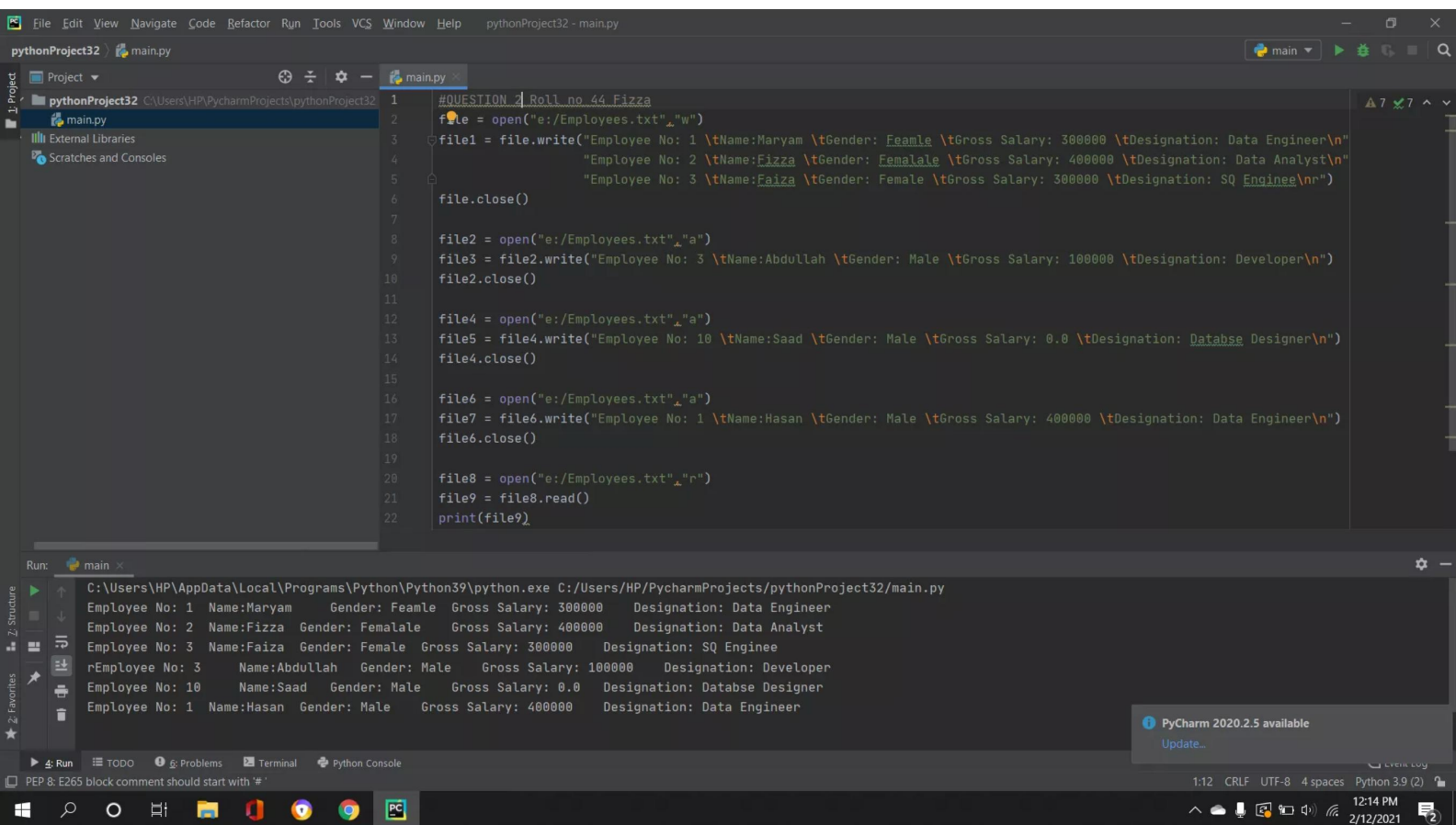
part b:-

```
def hit(n1, n2, c, x, y):  
    import math  
    d = math.sqrt(x**2 + y**2)  
    if d <= c:  
        print("TRUE")  
    else:
```

```
        print("FALSE")
```

```
n1 = int(input("Enter x-coordinate of centre:"))  
n2 = int(input("Enter y-coordinate of center:"))  
c = int(input("Enter radius:"))  
x = int(input("Enter x-coordinate of point P:"))  
y = int(input("Enter y-coordinate of point P:"))  
print(hit(n1, n2, c, x, y))
```





Employees.txt - Notepad

File Edit Format View Help

Employee No: 1	Name: Maryam	Gender: Feamle	Gross Salary: 300000	Designation: Data Engineer
Employee No: 2	Name: Fizza	Gender: Femalale	Gross Salary: 400000	Designation: Data Analyst
Employee No: 3	Name: Faiza	Gender: Female	Gross Salary: 300000	Designation: SQ Enginee
Employee No: 3	Name: Abdullah	Gender: Male	Gross Salary: 100000	Designation: Developer
Employee No: 10	Name: Saad	Gender: Male	Gross Salary: 0.0	Designation: Databse Designer
Employee No: 1	Name: Hasan	Gender: Male	Gross Salary: 400000	Designation: Data Engineer

Ln 1, Col 1 100% Windows (CRLF) UTF-8

14 items 1 item selected 557 bytes

Question No 3 :- part a:-

Functions :-

Function is a block of organized reusable code that is used to perform a single, related action. You use function in programming to bundle a set of instructions that you want to use repeatedly or that, because of their complexity are better self-contained in a sub-program and called when needed.

Recursive function:-

A recursive function is a function that calls itself during its execution. The process may repeat several times, outputting the result and the end of each iteration.

For Example:-

Function Count() uses recursion to count from any number between 9 to the number 10. Such as Count(6) would return 7, 8, 9, 10.

Random

numbers module. The built-in pseudo-random used to produce such as

For Example

(1) random
It will return a number from 0.0 to 1.0

(2) random
It will return a number between 0 and 1

part b:-

Random Module:-

You can generate random numbers in python by using random module. The random module is a built-in module to generate the pseudo-random variables. It can be used to perform some action randomly such as to get a random number.

For Examples:-

(1) `random.random()`

It will return a random number from 0.0 to 1.0.

(2) `random.randint(1, 50)`

It will return a random integer between the specified integers.

part C :-

Global and Local variables :-

Local variable is declared inside a function whereas Global variable is declared outside the function.

Local variables are related when the function has started execution and is lost when the function terminates, on the other hand Global variable is created as execution starts and is lost when the program ends.

part d :-

The sets, unlike list or tuples, cannot have multiple occurrences of the same element and store unordered values.

The both list and tuples are data structures in python but list are mutable and tuples are immutable.

List and tuples is an ordered collection of items whereas Dictionary is unordered collection and dictionary objects are mutable.

Part

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Example

General

Program

try:

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except

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• IO Ex

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Question No 4:-

Part b:-

Code

```
purchased = set()
price = set()
my_products = {'jam', 'peanut butter', 'breads',
               'chips', 'chocolates'}
x = int(input("How Many Items You Want To Purchase?"))
```

```
for i in range(x):
    a = input("Enter Your Item: ")
    if a in my_products:
        my_products.remove(a)
        purchased.add(a)
        b = int(input("The Price: "))
        price.add(b)
    else:
```

```
        print("Sorry your requirement is out of stock")
```

```
        continue.
```

```
print("Total items that you have purchased are:", len(purchased))
print("Your Bill is", sum(price), "rupees")
print("The max amount is", max(price), "and the min amount is", min(price))
```

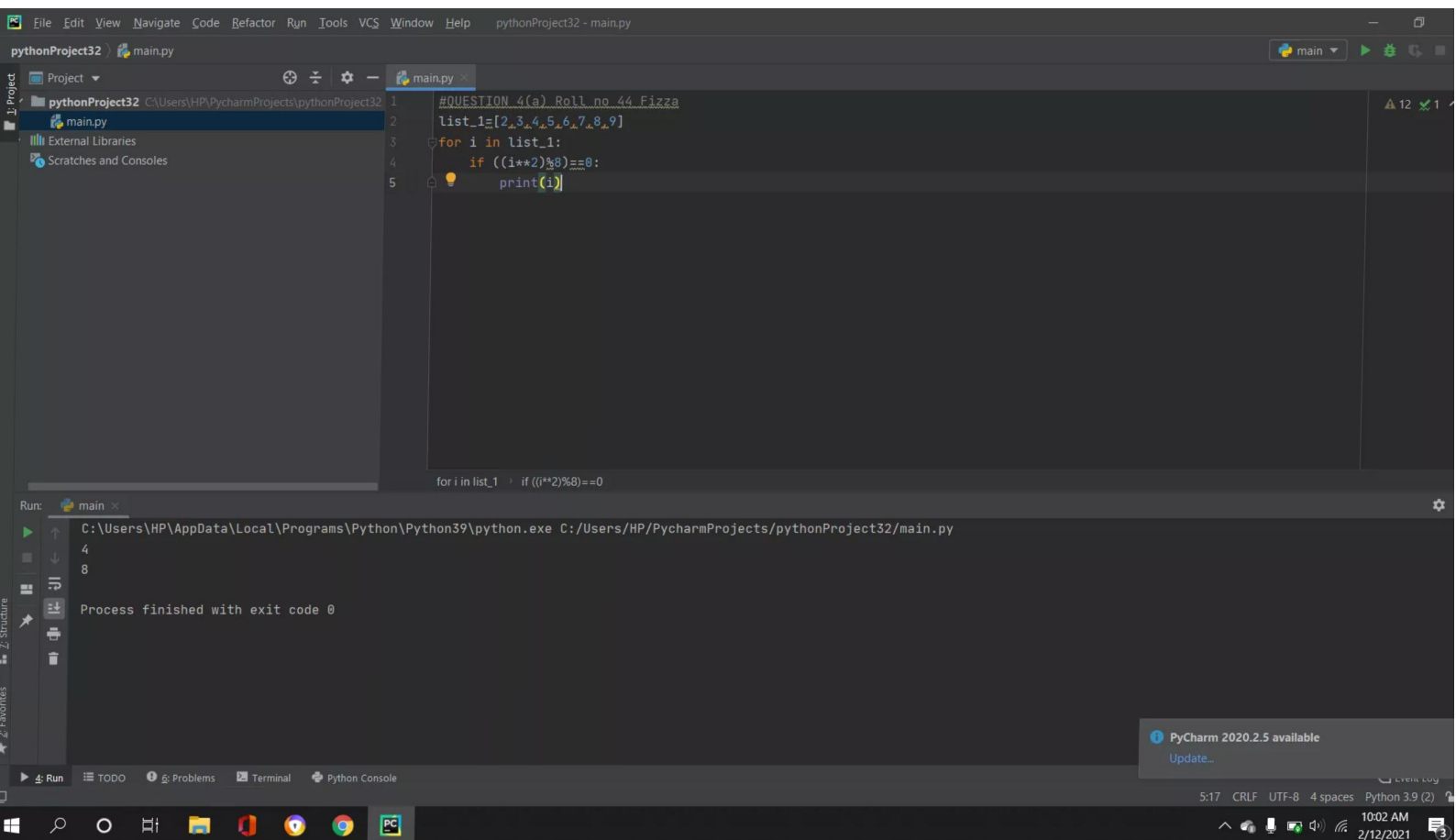

part ^a :-

```
list_1 = [2, 3, 4, 5, 6, 7, 8, 9]
```

```
for i in list_1:
```

```
    if (i**2)%8 == 0:
```

```
        print(i)
```

pythonProject32 - main.py

Project

pythonProject32 C:\Users\HP\PycharmProjects\pythonProject32

main.py

External Libraries

Scratches and Consoles

```
1 #QUESTION 4(b) Roll no 44 Fizza
2 purchased=set()
3 price=set()
4 my_products={'jam','peanut butter','breads','chips','chocolates'}
5 x=int(input("How Many Items You Want To Purchase ? "))
6 for i in range(x):
7     a=input("Enter Your Item : ")
8     if a in my_products:
9         my_products.remove(a)
10        purchased.add(a)
11        b=int(input("The Price : "))
12        price.add(b)
13    else:
14        print("Sorry your requirement is out of stock")
15        continue
16 print("Total items that you have purchased are : ",purchased)
17 print("Your Bill is",sum(price),"rupees")
18 print("The max amount is",max(price),"and the min amount is",min(price))
```

Run: main

C:\Users\HP\AppData\Local\Programs\Python\Python39\python.exe C:\Users\HP\PycharmProjects\pythonProject32/main.py

How Many Items You Want To Purchase ? 2

Enter Your Item : jam

The Price : 220

Enter Your Item : breads

The Price : 70

Total items that you have purchased are : {'jam', 'breads'}

Your Bill is 290 rupees

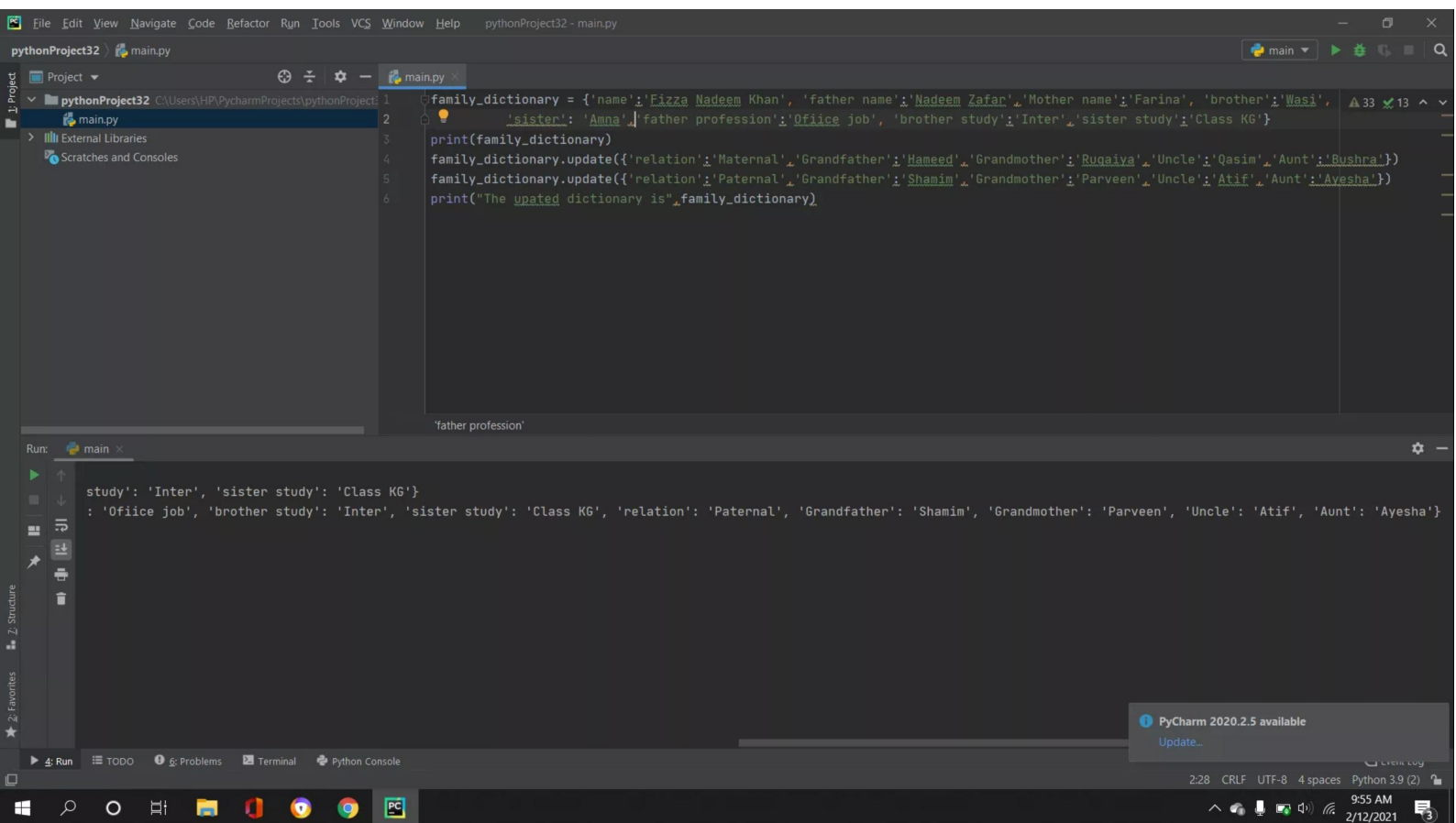
The max amount is 220 and the min amount is 70

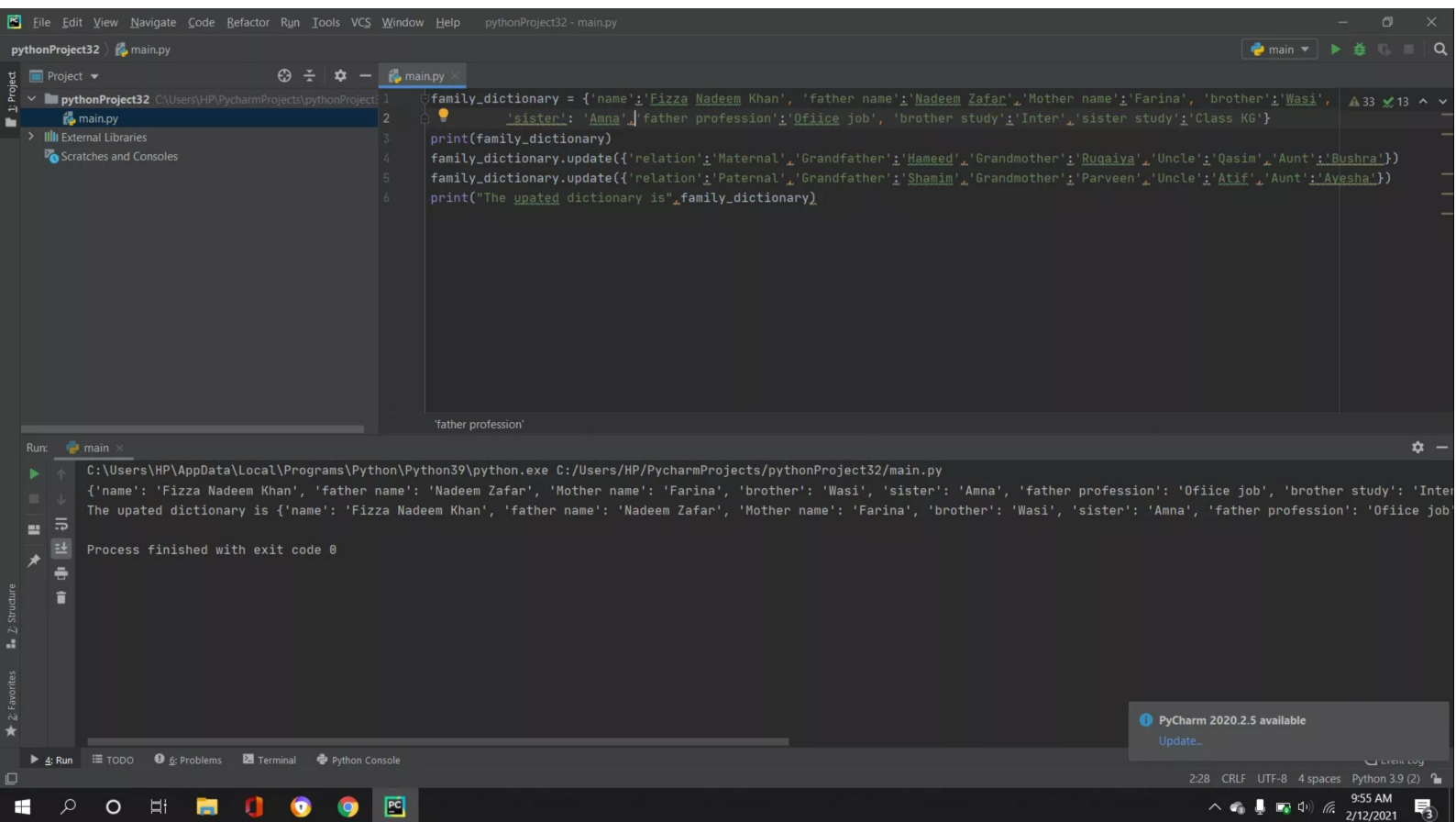
Process finished with exit code 0

PyCharm 2020.2.5 available
[Update...](#)

12:1 CRLF UTF-8 4 spaces Python 3.9 (2)

10:00 AM
2/12/2021





Question No 6:-

d')

```
from tkinter import *
```

```
x = Tk()
```

```
eel = IntVar()
```

```
def fact():
```

```
    n = eel.get()
```

```
    f = 1
```

```
    for i in range(n, 1, -1):
```

```
        f = f * i
```

```
    l3.config(text = f)
```

```
    return f
```

```
def even_odd():
```

```
    n = eel.get()
```

```
    if n % 2 == 0:
```

```
        l3.config(text = "Even")
```

```
    if n % 2 != 0:
```

```
        l3.config(text = "Odd")
```

```
    return
```

```
l1 = Label(x, width = 10, text = 'ENTER NO', bg = 'red')
```

```
l1.grid(row = 0, column = 0)
```

```
e1 = Entry(x, width = 10, bg = 'white', textvariable = eel)
```

```
e1.grid(row = 0, column = 1)
```

```
b1 = Button(x, width = 25, text = 'CALCULATE FACTORIAL',  
            bg = 'Yellow', command = fact)
```



```
b1.grid(row=2, column=1)
l2 = Label(x, width=15, text='RESULT', bg='red')
l2.grid(row=1, column=0)
l3 = Label(x, width=10, bg='White')
l3.grid(row=1, column=1)
b2 = Button(x, width=25, text='(CHECK EVEN OR ODD)',
bg='Yellow', command=even_odd)
b2.grid(row=3, column=1)
x.mainloop()
```

